PROXIANT ACADEMY

Course Catalogue

AI FUNDAMENTALS

A Comprehensive 4-Week Intensive Program

Curriculum Informed by Leading Programs at MIT, Stanford, and Carnegie Mellon University

2026 Edition

About Proxiant Academy

Proxiant Academy delivers industry-leading artificial intelligence education designed for professionals, aspiring practitioners, and curious minds. Our curriculum draws from the pedagogical frameworks of the world's top computer science programs including MIT's 6.034 Artificial Intelligence, Stanford's CS221 Principles and Techniques, and Carnegie Mellon University's BS in AI program.

We believe AI literacy is essential for every professional in the modern economy. Our courses combine rigorous theoretical foundations with hands-on practical applications, ensuring graduates are prepared to both understand and apply AI technologies in real-world contexts.

Course Overview: AI Fundamentals

Course Title	AI Fundamentals
Duration	4 Weeks (Intensive)
Format	Lectures, Hands-on Exercises, Weekly Projects, Assignments
Level	Foundational to Intermediate
Prerequisites	None. Basic Python familiarity and high-school mathematics are recommended but not required.
Certification	Proxiant Academy PCAP Certificate upon passing final assessment
Weekly Commitment	10–15 hours (lectures, exercises, projects, self-study)
Enrollment Fee	$1,499

Target Audience

This course is designed for a mixed audience spanning technical and non-technical backgrounds:

- Business leaders and managers seeking to understand AI capabilities and strategic implications
- Software developers and engineers looking to transition into AI/ML roles
- Data analysts wanting to expand their skill set into machine learning
- Students and recent graduates pursuing careers in artificial intelligence
- Entrepreneurs exploring AI-driven product opportunities
- Anyone with intellectual curiosity about how AI systems work

Learning Objectives

Upon successful completion of this course, students will be able to:

1. Define artificial intelligence, machine learning, and deep learning, and articulate how they relate to one another
2. Explain the historical development of AI from its origins to modern breakthroughs
3. Identify and describe core AI problem-solving methods including search algorithms, optimization, and constraint satisfaction
4. Distinguish between supervised, unsupervised, and reinforcement learning paradigms
5. Understand neural network architectures including CNNs, RNNs, and Transformers
6. Evaluate AI models using appropriate metrics and validation techniques
7. Describe real-world AI applications across industries including healthcare, finance, transportation, and creative arts
8. Understand the capabilities and limitations of Large Language Models and Generative AI
9. Analyze ethical implications of AI systems including bias, fairness, transparency, and accountability
10. Develop a strategic perspective on AI adoption and implementation in organizational contexts

Course Structure: Weekly Breakdown

Week 1: Foundations of Artificial Intelligence

This opening week establishes the conceptual foundations of AI, tracing its history from Turing's seminal work to today's breakthroughs. Students explore what intelligence means computationally, examine the major paradigms of AI, and learn foundational problem-solving techniques including search and knowledge representation. Inspired by the opening modules of MIT 6.034 and Stanford CS221.

Topics Covered:

- What is Artificial Intelligence? Definitions, scope, and the Turing Test
- History of AI: From Dartmouth Conference (1956) to modern deep learning
- Types of AI: Narrow AI, General AI, and Superintelligence concepts
- AI vs. Machine Learning vs. Deep Learning vs. Data Science
- Problem solving as search: State spaces, goal trees, and search strategies
- Uninformed search: Breadth-first, depth-first, uniform-cost search
- Informed search: Heuristics, greedy best-first search, A* algorithm
- Knowledge representation: Logic, semantic networks, and ontologies
- Intelligent agents: Perception, reasoning, and action
- Hands-on: Implementing basic search algorithms and exploring AI tools

Week 2: Machine Learning Fundamentals

Week 2 dives into the core of modern AI: machine learning. Students learn how machines learn from data through supervised, unsupervised, and reinforcement learning paradigms. Drawing from Stanford CS221's approach to learning theory and CMU's emphasis on practical application, this week builds both theoretical understanding and hands-on skills.

Topics Covered:

- Introduction to Machine Learning: Learning from data, the ML pipeline
- Supervised Learning: Regression and classification frameworks
- Linear models: Linear regression, logistic regression, and gradient descent
- Decision trees and ensemble methods: Random forests, boosting
- Support Vector Machines: Maximum margin classifiers and kernel methods
- Unsupervised Learning: Clustering (K-means, hierarchical), dimensionality reduction (PCA)
- Model evaluation: Training/validation/test splits, cross-validation, bias-variance tradeoff
- Performance metrics: Accuracy, precision, recall, F1-score, ROC-AUC
- Feature engineering: Selection, extraction, and transformation techniques
- Introduction to Reinforcement Learning: Agents, environments, rewards, and policies
- Hands-on: Building ML models with real-world datasets

Week 3: Deep Learning and Neural Networks

This week explores the deep learning revolution that has transformed AI capabilities. Students study neural network architectures from foundational perceptrons to modern Transformers, understanding how these systems learn hierarchical representations. Aligned with CMU's AI Engineering certificate and Stanford's deep learning modules.

Topics Covered:

- Neural network fundamentals: Perceptrons, activation functions, backpropagation
- Multi-layer networks: Architecture design, hidden layers, and universal approximation
- Training deep networks: Optimization (SGD, Adam), regularization (dropout, batch normalization)
- Convolutional Neural Networks (CNNs): Architecture, convolution operations, pooling, and applications in computer vision
- Recurrent Neural Networks (RNNs): Sequential data processing, LSTMs, and GRUs
- The Transformer architecture: Self-attention mechanism, encoder-decoder structure
- Transfer learning: Pre-trained models, fine-tuning strategies
- Computer Vision applications: Image classification, object detection, segmentation
- Natural Language Processing fundamentals: Tokenization, embeddings, language models
- Hands-on: Building and training neural networks, experimenting with pre-trained models

Week 4: Generative AI, Ethics, and AI Strategy

The final week brings everything together with a focus on cutting-edge generative AI, responsible AI practices, and strategic implementation. Students explore Large Language Models, AI ethics frameworks aligned with CMU's emphasis on societal impact, and develop a roadmap for applying AI in their professional contexts.

Topics Covered:

- Large Language Models (LLMs): GPT, Claude, and the scaling paradigm
- Generative AI: Text generation, image synthesis (diffusion models), and multimodal AI
- Prompt engineering: Techniques for effective interaction with AI systems
- AI Ethics and Responsible AI: Bias, fairness, transparency, and accountability
- AI Safety: Alignment, hallucinations, and reliability challenges
- Privacy and data governance in AI systems
- Regulatory landscape: EU AI Act, NIST AI Framework, and global perspectives
- AI in industry: Healthcare, finance, manufacturing, creative industries, and more
- AI Strategy: Building an AI roadmap, organizational readiness, and change management
- The future of AI: AGI debate, emerging trends, and career opportunities
- Capstone review and certification preparation

Assessment and Certification

Students are assessed through a combination of continuous evaluation and a final certification exam:

Component	Weight	Description
Weekly Exercises	15%	In-class and take-home exercises for each week
Weekly Assignments	20%	Graded homework assignments due end of each week
Weekly Projects	30%	Hands-on projects applying concepts to real scenarios
Final Certification Exam	35%	Comprehensive exam covering all 4 weeks of material

Passing Score: 70% overall weighted average required for certification.

Certificate: Proxiant Academy PCAP (Proxiant Certified AI Professional) Certificate, verifiable through our online credential system.

PCAP Certification Details

Upon successful completion of the AI Fundamentals course, you become eligible for the Proxiant Certified AI Professional (PCAP) exam:

Exam Name	Proxiant Certified AI Professional (PCAP)
Exam Duration	3 hours
Question Format	100 Multiple Choice Questions (MCQs)
Passing Score	70% (70 out of 100 questions)
Exam Cost	$199
Certificate Validity	3 years from date of certification
Credential Verification	Digital badge + verifiable online credential

Course Materials Provided

- Lecture slide decks for all 4 weeks (PPTX format)
- Comprehensive class notes for each session
- In-class and take-home exercises with detailed solutions
- Weekly graded assignments with solutions
- Weekly hands-on projects with solution guides
- Final certification exam with answer key
- PCAP exam preparation materials and practice tests
- Supplementary reading list and resource guide

Why Proxiant Academy?

- Curriculum benchmarked against MIT, Stanford, and Carnegie Mellon University programs
- Balanced approach for both technical and non-technical learners
- Hands-on projects using real-world datasets and industry tools
- Small cohort sizes for personalized attention and mentorship
- Industry-recognized PCAP certification upon successful completion
- Access to alumni network and ongoing learning resources

Enroll today at proxiant.ai/academy